

动态 Fee 参数详细报告 (r80 方案)

组合: f2-v3000-r80-d14 | 目标: 在可控尖峰下提升时间敏感性

结论概览:

- 1) 当前组合在本次样本下 >20% 尖峰为 0, >10% 尖峰为 0。
- 2) 时间区分度: dMean(0~2 vs 2~10)=0.1589%, dMean(2~10 vs 10~30)=0.0399%。
- 3) 对比 r85-d14: r80 更平滑, 但时间区分度略弱。

区间	平均动态费占比	P95	平均动态费(USDB)	平均成交额(USDB)
0to2s	0.4026%	0.6453%	0.3885	89.9717
2to10s	0.2438%	0.3712%	0.3248	89.9717
10to30s	0.2039%	0.3594%	0.3065	89.9717

测试方法与环境

Methodology / Environment

测试方法:

- 使用脚本: test/trade/random-yes-trades-1000.spec.js
- 每组场景执行随机买卖, 统计 SELL 侧动态费占比
- 动态费占比定义: $\text{dynamicFeeUSDB} / \text{grossOutUSDB} * 100\%$
- 同时输出原始曲线与 P95 过滤曲线

测试环境:

- Hardhat 本地网络 (chain simulation)
- 部署方式: deployCleanFixture()
- 交易笔数: 200/场景
- 场景矩阵: 3 个时间区间 x 3 个交易日区间
时间: 0~2s, 2~10s, 10~30s
交易额: 5~100U, 100~200U, 200~300U

测试参数与口径

Parameters / Metrics

参数	取值	说明
filterPeriod	2	高频过滤窗口(秒)
decayPeriod	14	波动衰减窗口(秒)
reductionFactor	800000000000000000	衰减比例(1e18 精度)
variableFeeControl	3000000000000000	波动费强度因子
maxAccumulator	1000000000	波动累积上限
baseFeeUnit	2000000000000	按 tickCrossed 的基础费单元

指标	定义
avg_ratio	平均动态费占比(%)
p95_ratio	动态费占比 95 分位(%)
gt10 / gt20	动态费占比 >10% / >20% 的笔数
dMean(0~2 vs 2~10)	同交易额档内，时间区间均值差
dMean(2~10 vs 10~30)	同交易额档内，时间区间均值差(重点)

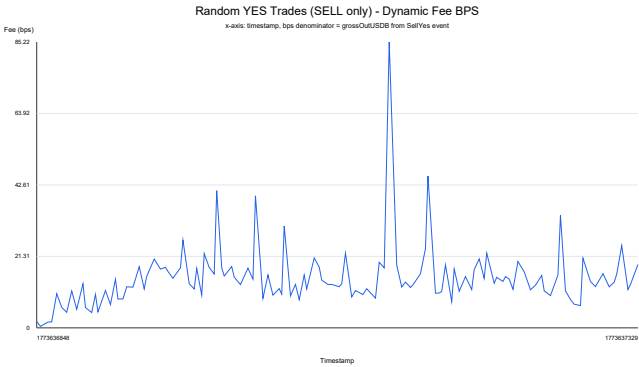
场景统计总表

9 Scenarios Summary

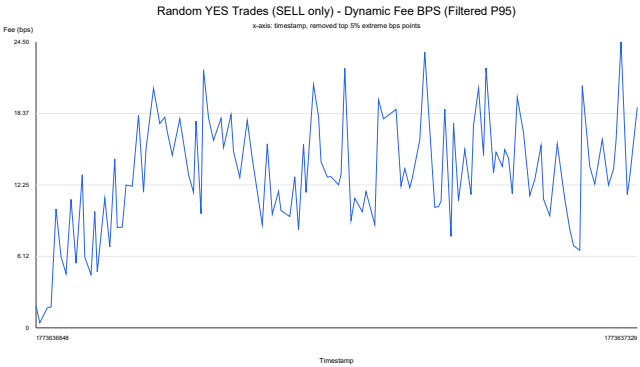
场景	SELL数	平均成交额	平均动态费占比	P95	最大	>10%	>20%
0to2s/5to100U	126	32.17	0.149%	0.245%	0.852%	0	0
0to2s/100to200U	124	91.34	0.463%	0.569%	8.707%	0	0
0to2s/200to300U	126	146.40	0.596%	1.122%	3.129%	0	0
2to10s/5to100U	126	32.17	0.087%	0.177%	0.214%	0	0
2to10s/100to200U	124	91.34	0.249%	0.373%	0.862%	0	0
2to10s/200to300U	126	146.40	0.396%	0.563%	0.857%	0	0
10to30s/5to100U	126	32.17	0.076%	0.169%	0.210%	0	0
10to30s/100to200U	124	91.34	0.208%	0.353%	0.397%	0	0
10to30s/200to300U	126	146.40	0.328%	0.556%	0.568%	0	0

0to2s/5to100U 动态费曲线对比

原始动态费占比曲线



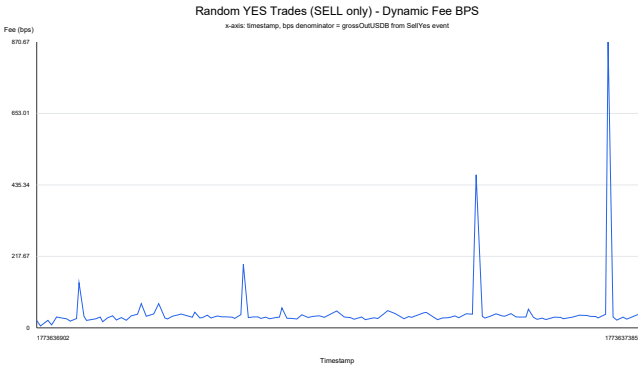
尖峰过滤后 (P95) 曲线



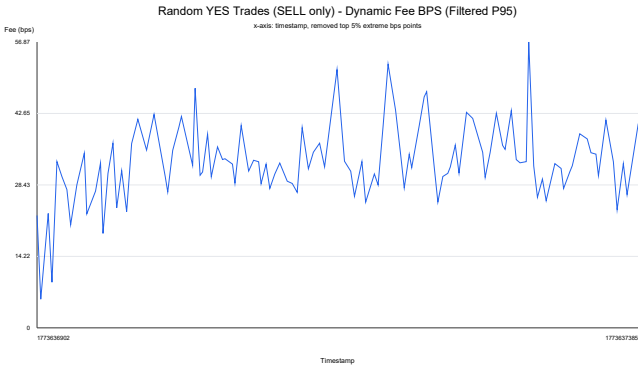
统计: SELL=126 | avg_ratio=0.1486% | p95=0.2450% | max=0.8522%
avg_trade=32.1667 USDB | avg_dynamic_fee=0.0485 USDB | >10%=0 | >20%=0

0to2s/100to200U 动态费曲线对比

原始动态费占比曲线

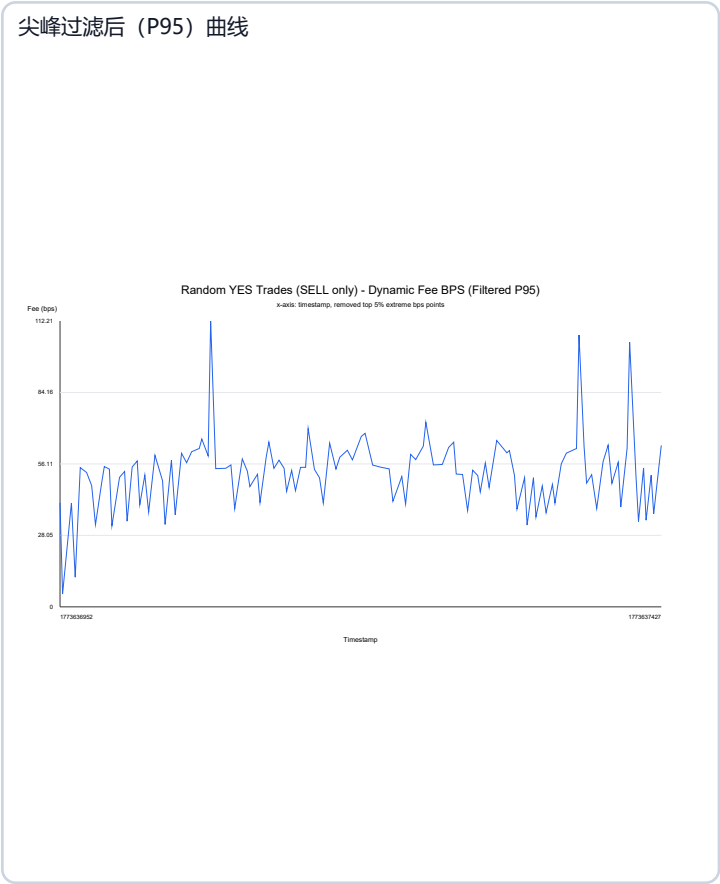
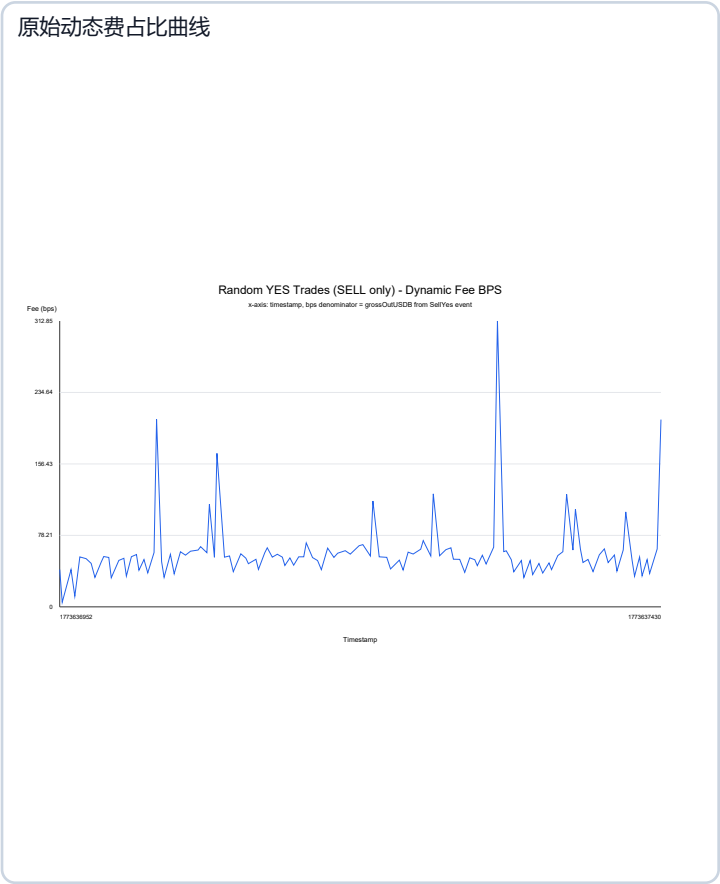


尖峰过滤后 (P95) 曲线



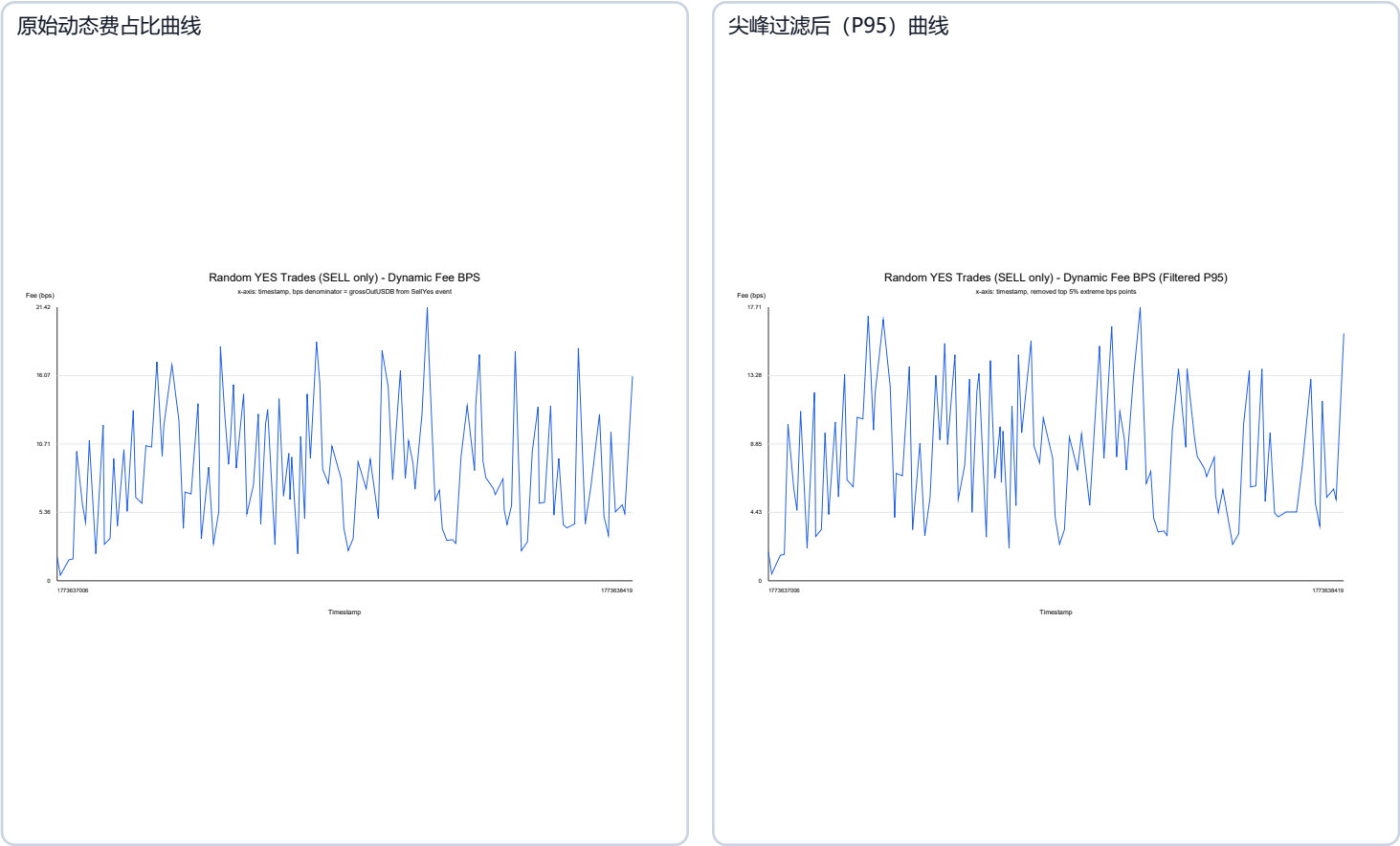
统计: SELL=124 | avg_ratio=0.4630% | p95=0.5687% | max=8.7067%
avg_trade=91.3438 USDB | avg_dynamic_fee=0.3160 USDB | >10%=0 | >20%=0

0to2s/200to300U 动态费曲线对比



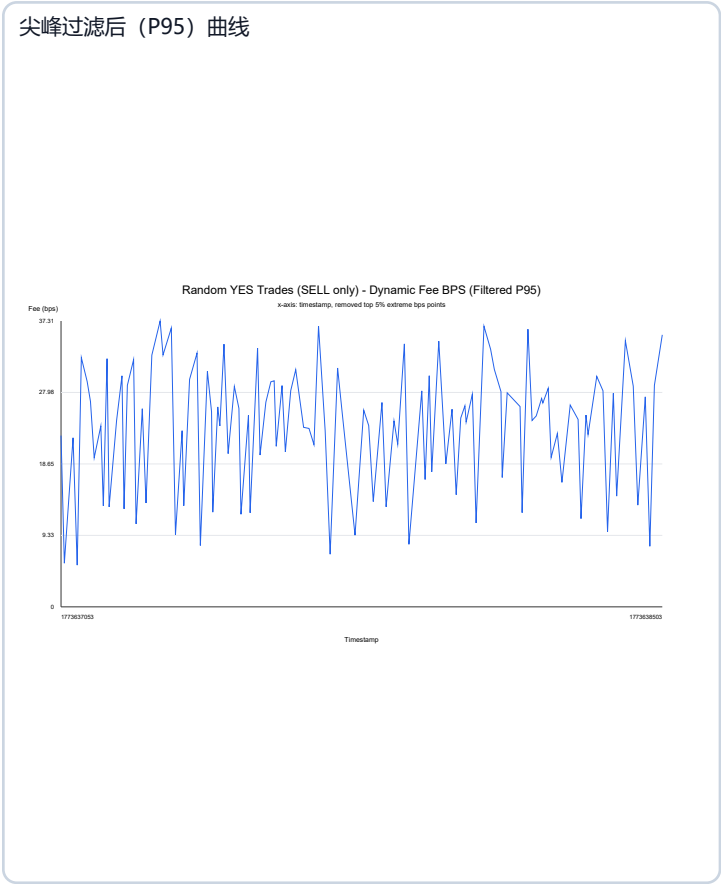
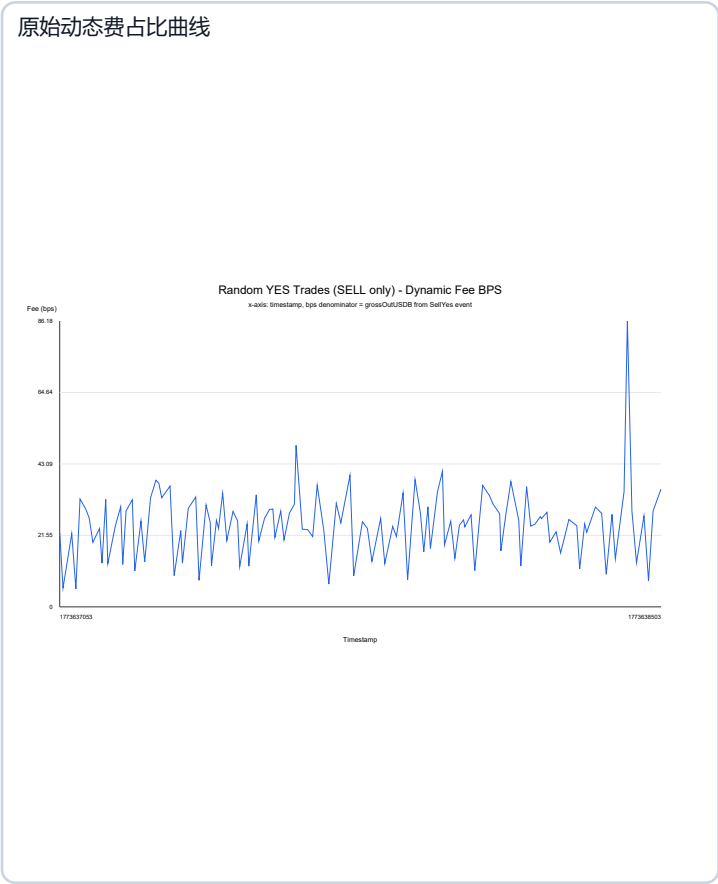
统计: SELL=126 | avg_ratio=0.5963% | p95=1.1221% | max=3.1285%
avg_trade=146.4048 USDB | avg_dynamic_fee=0.8011 USDB | >10%=0 | >20%=0

2to10s/5to100U 动态费曲线对比



统计: SELL=126 | avg_ratio=0.0870% | p95=0.1771% | max=0.2142%
avg_trade=32.1667 USDB | avg_dynamic_fee=0.0371 USDB | >10%=0 | >20%=0

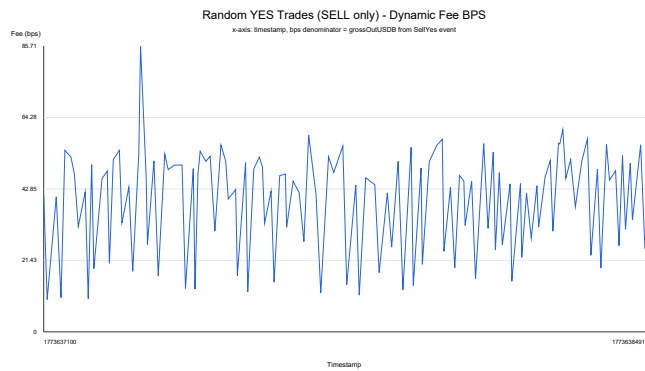
2to10s/100to200U 动态费曲线对比



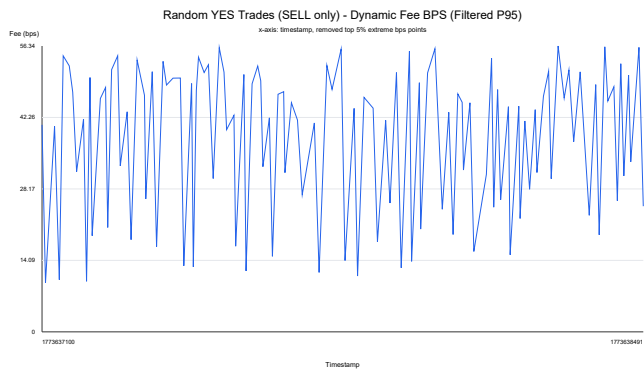
统计: SELL=124 | avg_ratio=0.2485% | p95=0.3731% | max=0.8618%
avg_trade=91.3438 USDB | avg_dynamic_fee=0.2547 USDB | >10%=0 | >20%=0

2to10s/200to300U 动态费曲线对比

原始动态费占比曲线

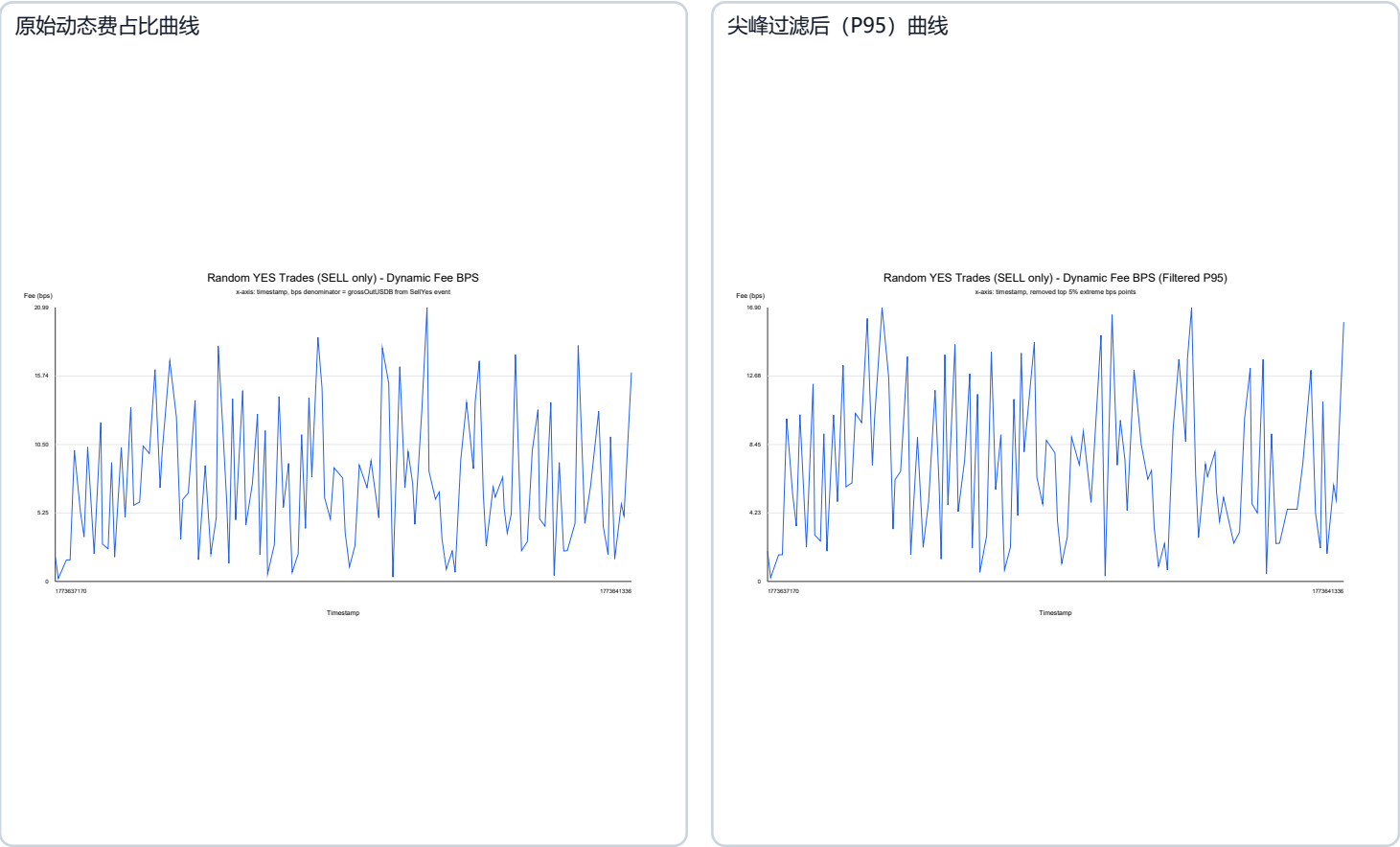


尖峰过滤后 (P95) 曲线



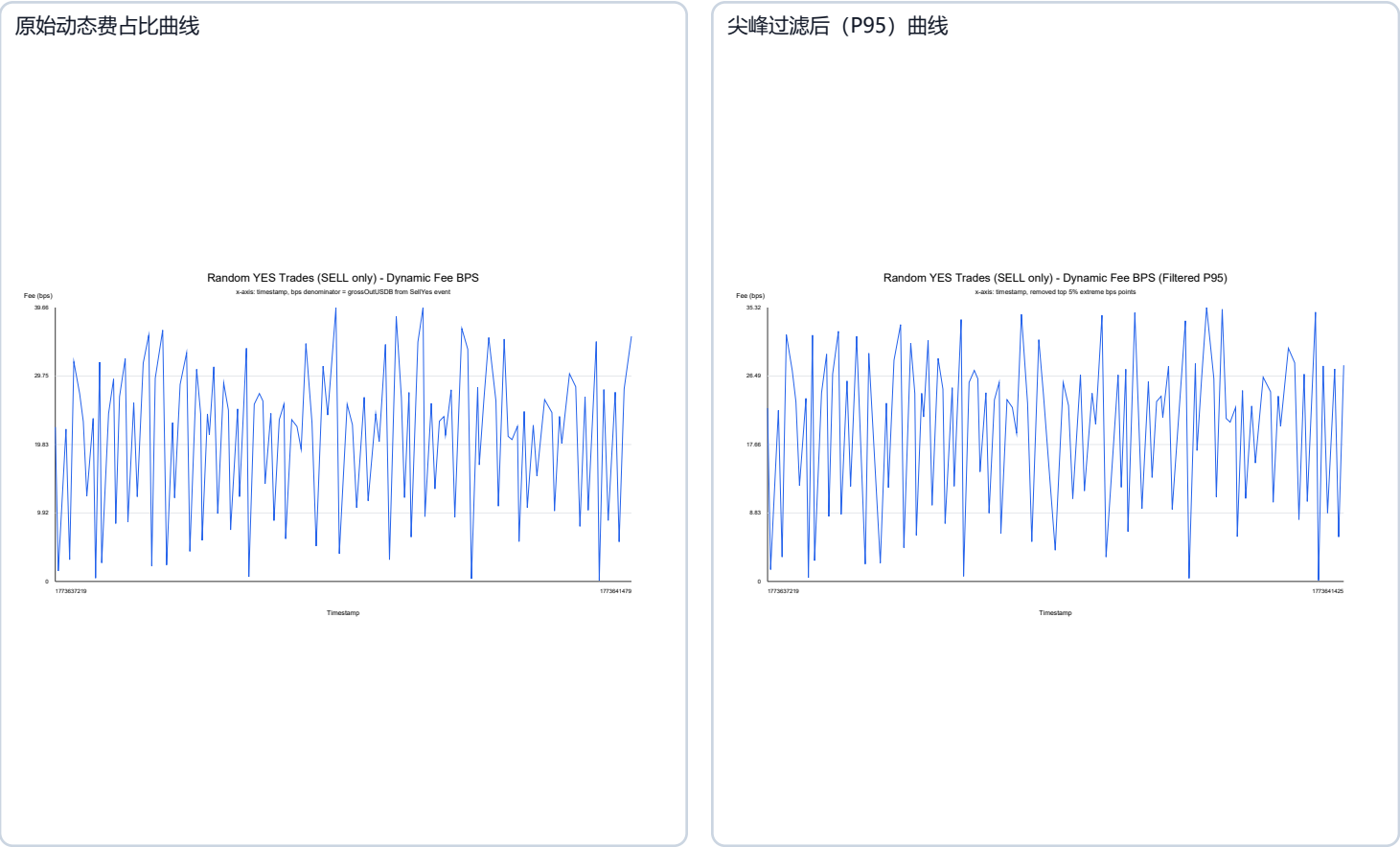
统计: SELL=126 | avg_ratio=0.3959% | p95=0.5634% | max=0.8571%
avg_trade=146.4048 USDB | avg_dynamic_fee=0.6825 USDB | >10%=0 | >20%=0

10to30s/5to100U 动态费曲线对比



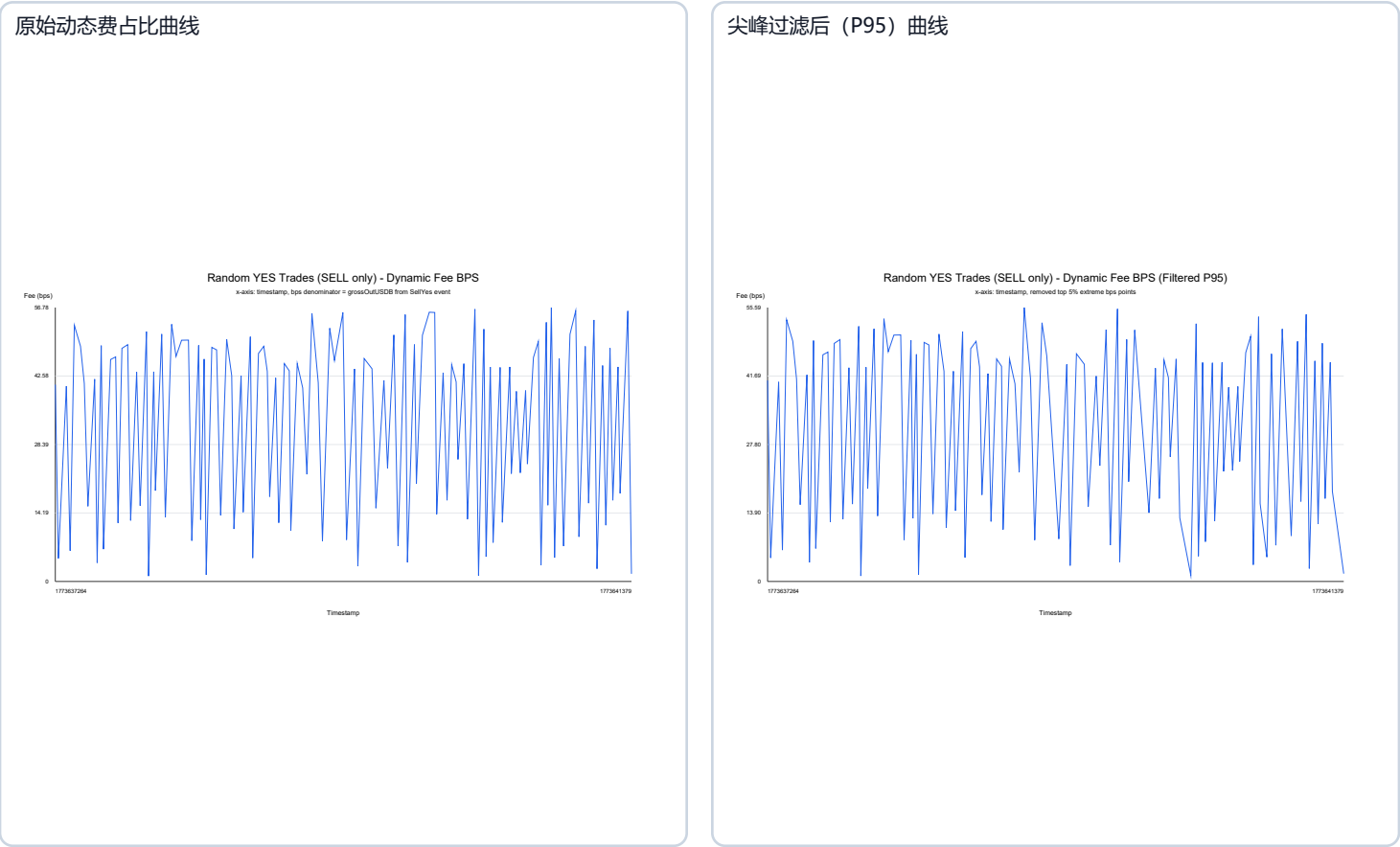
统计: SELL=126 | avg_ratio=0.0755% | p95=0.1690% | max=0.2099%
avg_trade=32.1667 USDB | avg_dynamic_fee=0.0353 USDB | >10%=0 | >20%=0

10to30s/100to200U 动态费曲线对比



统计: SELL=124 | avg_ratio=0.2084% | p95=0.3532% | max=0.3966%
avg_trade=91.3438 USDB | avg_dynamic_fee=0.2408 USDB | >10%=0 | >20%=0

10to30s/200to300U 动态费曲线对比



统计: SELL=126 | avg_ratio=0.3277% | p95=0.5559% | max=0.5678%
avg_trade=146.4048 USDB | avg_dynamic_fee=0.6434 USDB | >10%=0 | >20%=0